

CardioRisk Prediction

Model Selection Assignment — Report

Assignment ID: MS/01 | Total Marks: 150

This report presents the end-to-end analysis pipeline for cardiovascular disease risk prediction — covering data cleaning, EDA, feature engineering, model building, hyperparameter tuning, and final recommendations.

All source files (notebook, dataset, and report) are available at the GitHub repository:
<https://github.com/mayuresh-tungare/c6-assignment/tree/main>

1. Problem Statement and Objective

Business Context: CardioCare requires a machine learning model to predict cardiovascular disease (CVD) risk from patient health data, enabling clinical teams to prioritise high-risk patients for consultation.

Objectives:

- Analyse and preprocess patient physiological and lifestyle data
- Build and compare SVM and Decision Tree classifiers for binary CVD classification
- Optimise thresholds and hyperparameters to maximise recall for positive CVD cases
- Produce actionable insights and a deployable screening model

2. Dataset Overview

70,000 patient records with 14 features. The target variable **cardio** indicates CVD presence.

Column	Description	Type
age	Age (days, converted to years)	Numerical
gender	0=Female, 1=Male	Categorical
height	Height cm (converted to metres)	Numerical
weight	Weight in kg	Numerical
ap_hi	Systolic blood pressure	Numerical
ap_lo	Diastolic blood pressure	Numerical
cholesterol	0=Normal, 1=Above, 2=Well Above	Categorical
gluc	0=Normal, 1=Above, 2=Well Above	Categorical
smoke	0=No, 1=Yes	Binary
alco	0=No, 1=Yes	Binary
active	0=No, 1=Yes	Binary
cardio	CVD Target (0=No, 1=Yes)	Target

3. Assumptions

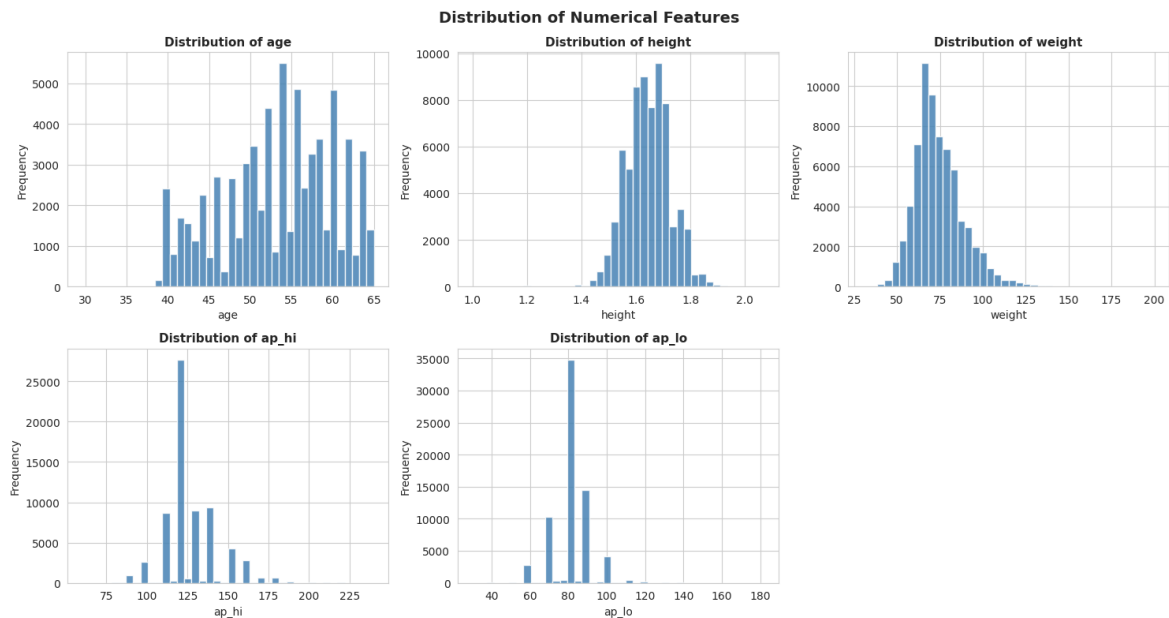
- Blood pressure values outside 30-300 mmHg removed (physiologically invalid)
- Diastolic (ap_lo) must not exceed systolic (ap_hi); violating rows dropped
- Height range 100-220 cm and weight 30-200 kg used as bounds
- Age stored in days converted to years; height cm to metres
- Cholesterol and glucose categories 1 and 2 merged into single elevated category
- BMI (weight/height²) replaces raw height and weight to reduce multicollinearity
- 70:30 stratified train-test split with random_state=42
- MinMaxScaler applied to numerical features (fitted on train only)
- 15,000-sample subsample used for SVM training ($O(n^2)$ computational constraint)
- Probability threshold 0.40 chosen to favour recall in a clinical screening context

4. Exploratory Data Analysis and Feature Engineering

4.1 Data Cleaning

Statistical summary revealed blood pressure extremes (ap_hi up to 16,020 mmHg) and negative values. After filtering, **1,353 rows (1.9%) were removed**, leaving 68,647 clean records.

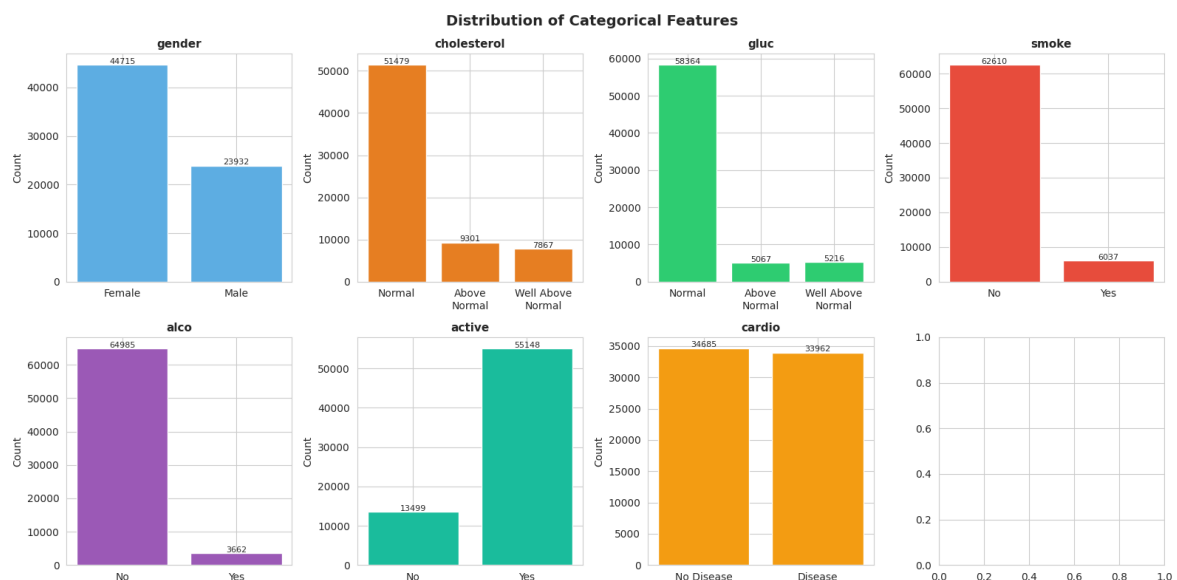
4.2 Univariate Analysis — Numerical Features



Distribution of numerical features after cleaning and unit conversion

- **Age:** Approx. normal, centred ~53 years (skew=-0.11). Range 29.6-65.0 yrs
- **Weight:** Right-skewed (skew=0.98); some high-BMI outliers visible
- **Blood pressure:** Slight right skew; most values within normal clinical ranges

4.3 Univariate Analysis — Categorical Features

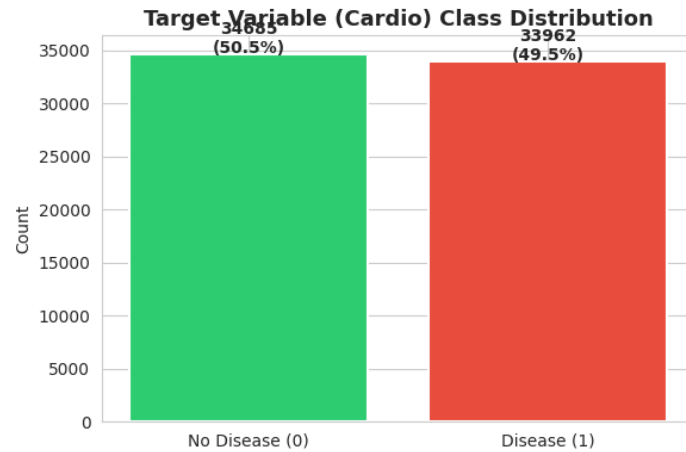


Distribution of all categorical features

- **Gender:** ~65% female, 35% male

- Cholesterol: 75% normal, 14% above normal, 11% well above normal
- Smoking (~9%) and alcohol (~5%): minority of patients
- Physical activity: ~80% patients are active

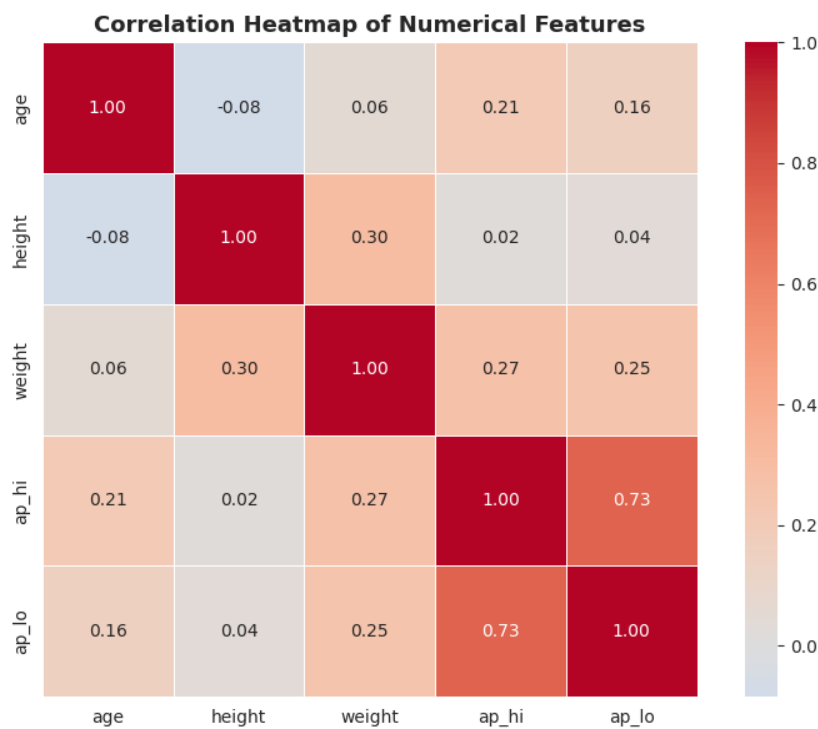
4.4 Target Variable Distribution



CVD target class balance

Dataset is **near-perfectly balanced** (50.5% no disease / 49.5% disease). No resampling required.

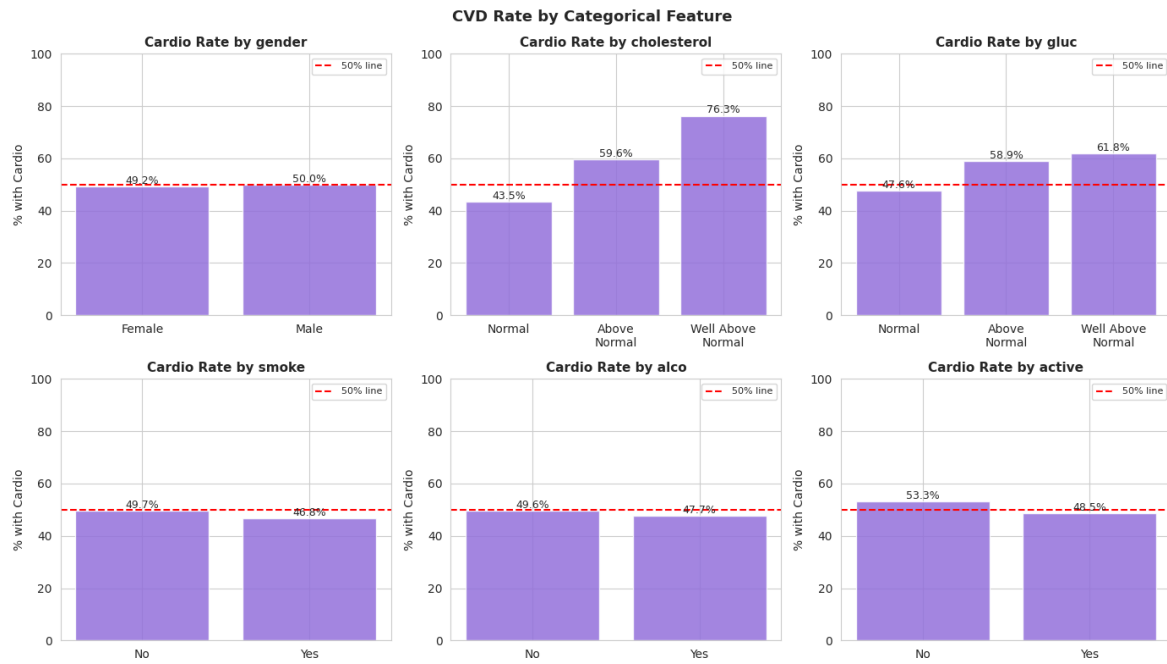
4.5 Correlation Analysis



Pearson correlation heatmap of numerical features

- **ap_hi vs ap_lo**: Strong correlation ($r=0.734$) — expected clinically
- **height vs weight**: Moderate ($r=0.302$) — motivation for BMI feature
- **age vs blood pressure**: Moderate ($r\sim0.21$) — BP increases with age

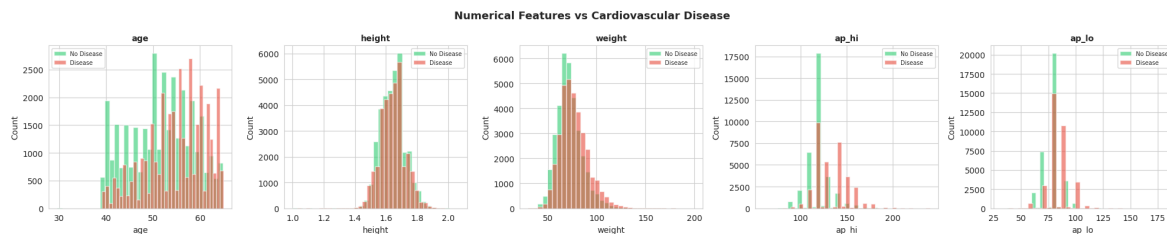
4.6 Bivariate Analysis — Categorical Features vs CVD



CVD prevalence rate (%) by categorical feature category

- **Cholesterol:** Strongest predictor — 76% CVD at well-above-normal vs 44% normal (32pp gap)
- **Glucose:** Moderate signal — 62% CVD elevated vs 48% normal
- **Physical activity:** Mildly protective — inactive: 53% vs active: 49%
- **Gender, smoking, alcohol:** Surprisingly weak predictors (near 50% for both categories)

4.7 Bivariate Analysis — Numerical Features vs CVD

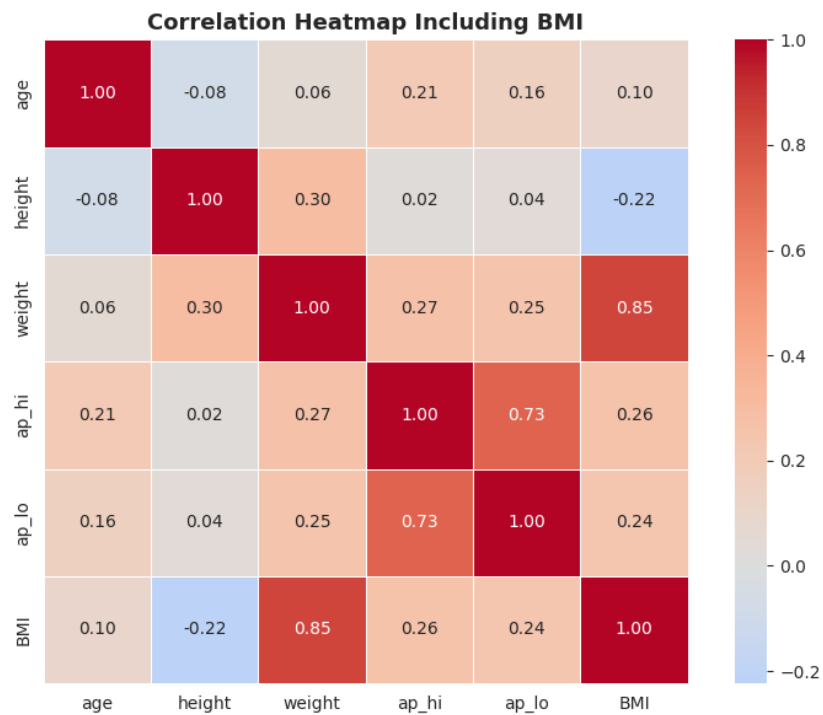


Distribution of numerical features by CVD status

- Blood pressure: clearest separation — CVD patients show higher ap_hi and ap_lo distributions
- Age: CVD patients are systematically older
- Height: minimal separation — weak predictor

4.8 Feature Engineering

4.8.1 BMI and Multicollinearity



Correlation heatmap including engineered BMI feature

BMI is highly correlated with weight ($r=0.856$). Both height and weight were **dropped** after BMI creation to eliminate multicollinearity.

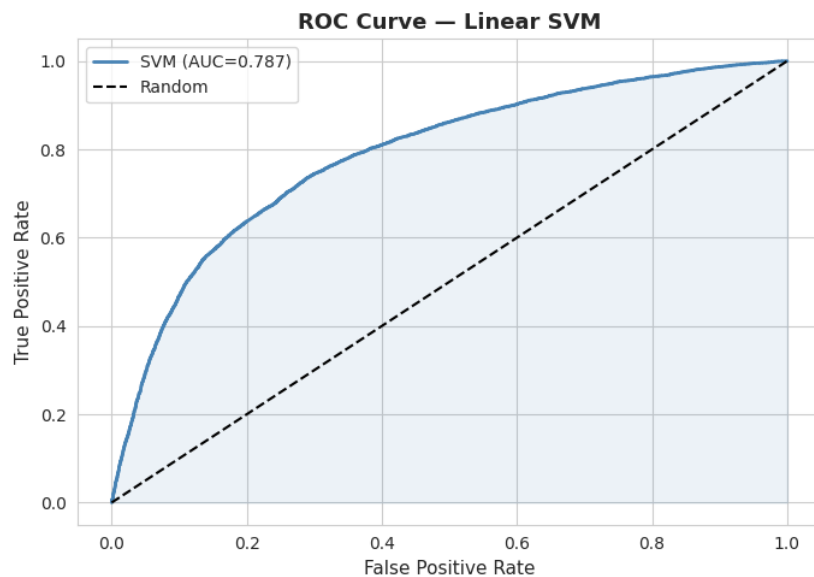
4.8.2 Category Combination and Final Features

Feature	Before	After Combining
cholesterol	0:35,945 1:6,522 2:5,585	0:35,945 1:12,107
gluc	0:40,797 1:3,606 2:3,649	0:40,797 1:7,255

Final 10 features: age, ap_hi, ap_lo, BMI (scaled), smoke, alco, active, gender_1, cholesterol_1, gluc_1

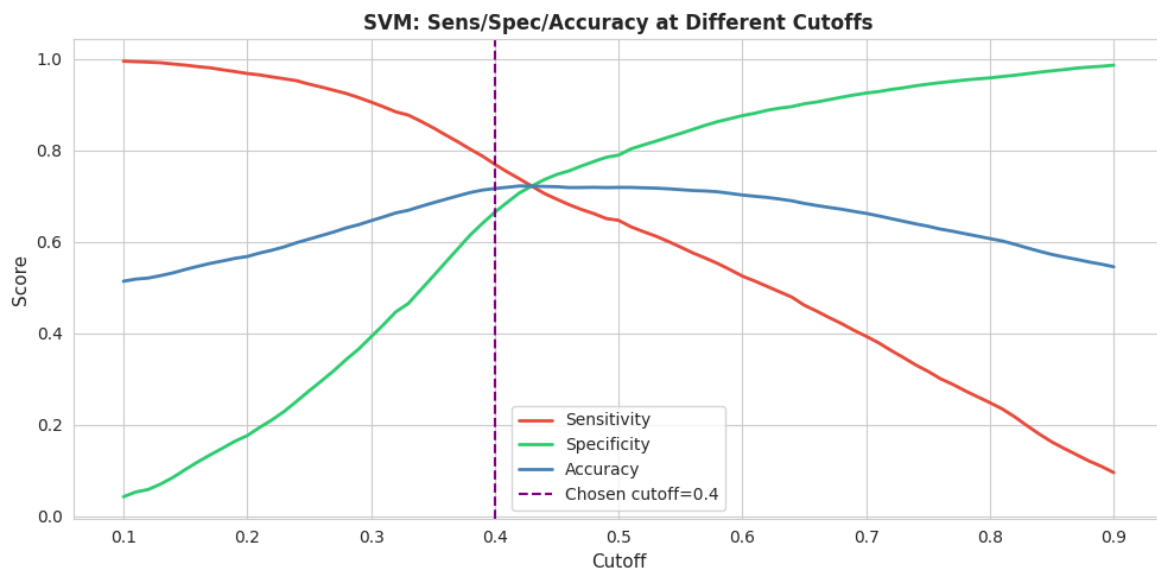
5. Model Building and Rationale

5.1 Linear SVM



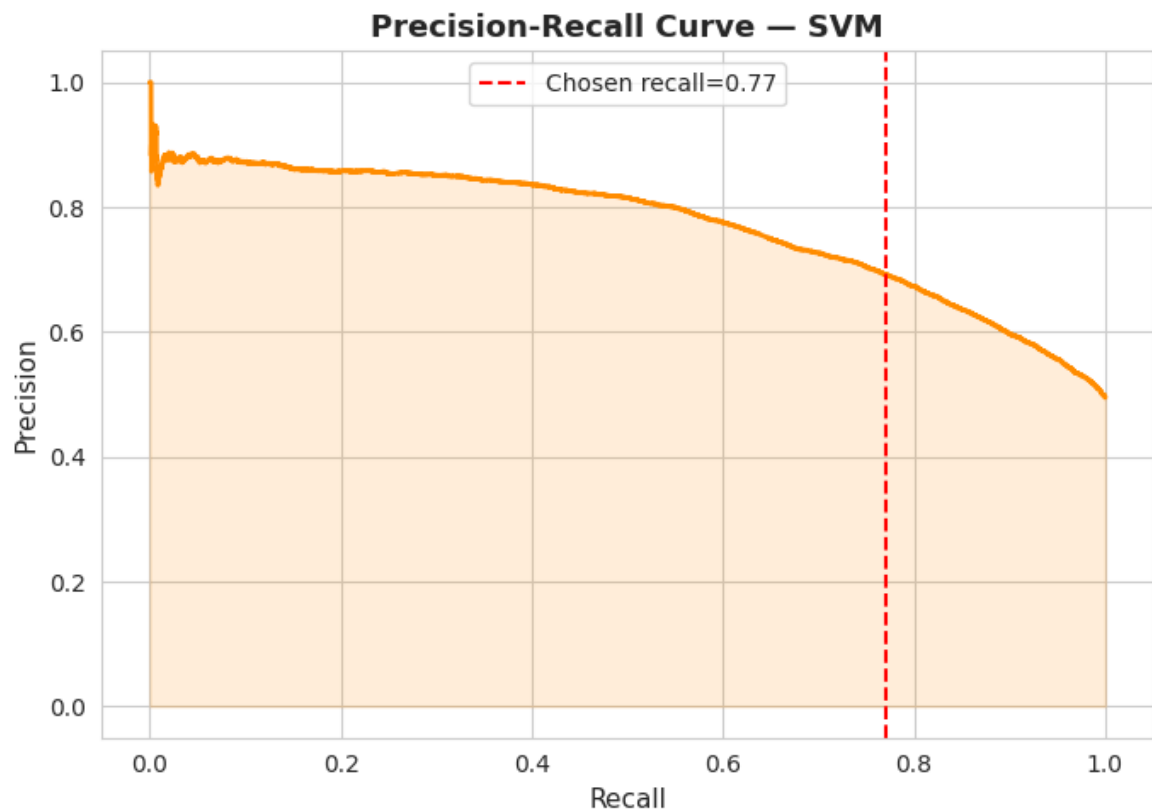
ROC Curve — Linear SVM (AUC = 0.787)

AUC = **0.787**: the SVM correctly ranks ~79% of positive-negative patient pairs. Linear kernel is appropriate as classes show substantial linear separability after scaling.



SVM: Sensitivity, Specificity, Accuracy across probability thresholds

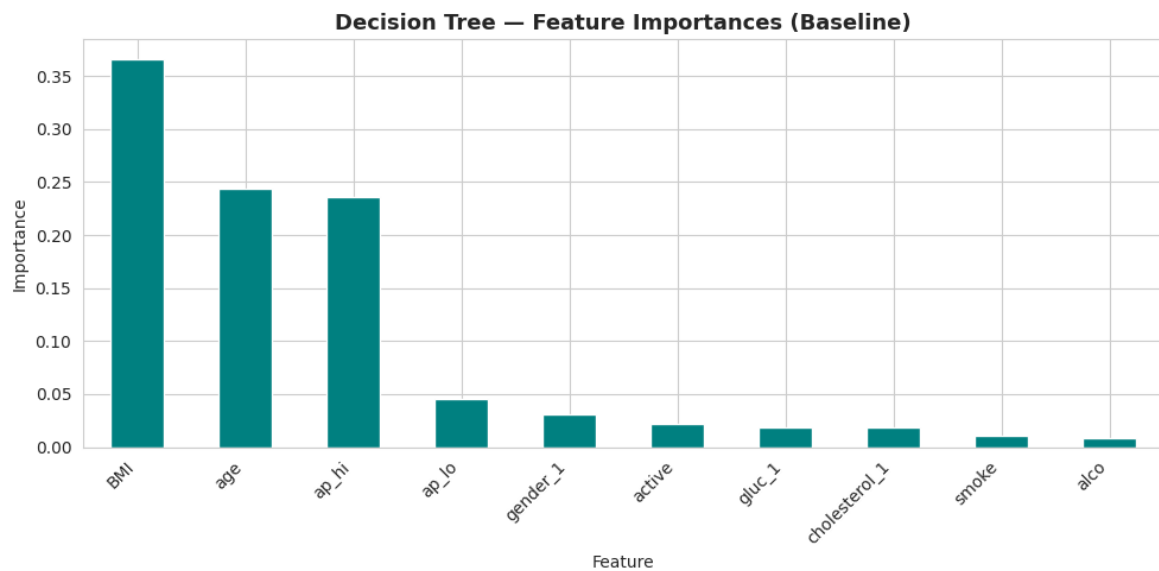
Sensitivity-specificity crossover ~0.43. Threshold **0.40** chosen to favour recall (catch more CVD-positive patients) over specificity.



Precision-Recall Curve — Linear SVM

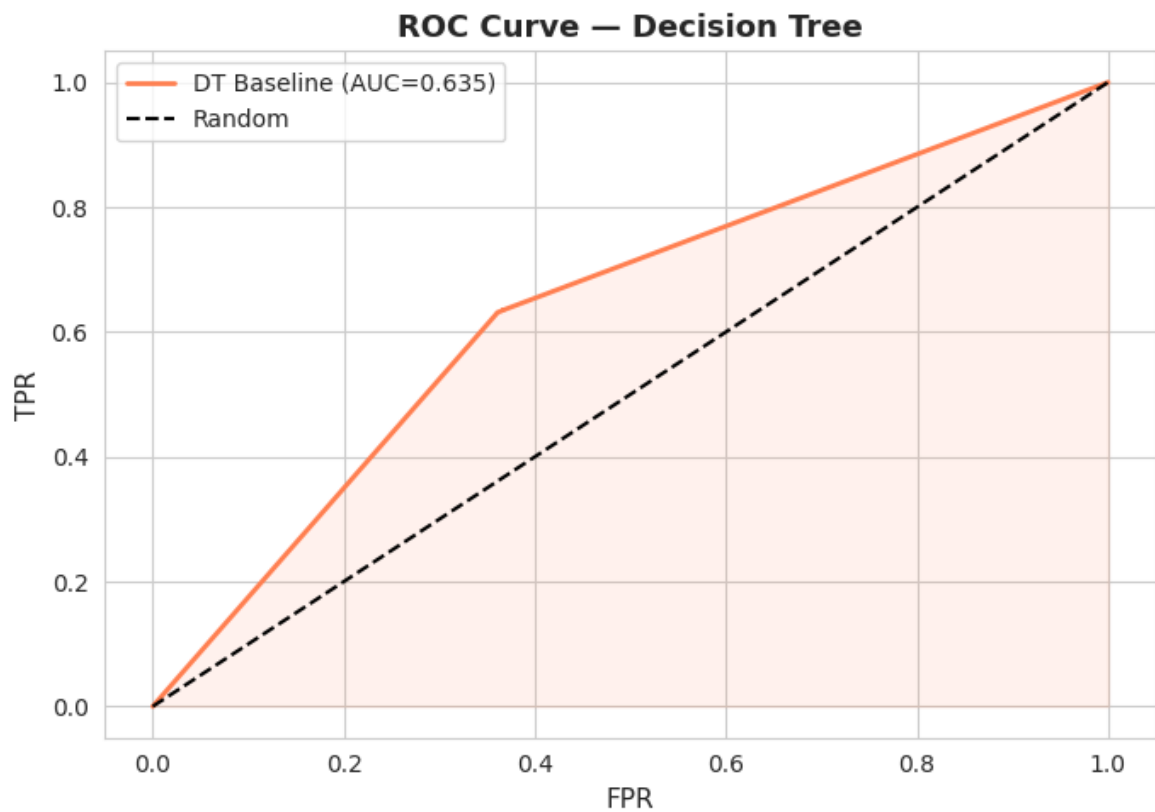
At threshold 0.40: Recall = **0.77**, Precision = **0.69**.

5.2 Decision Tree (Baseline and Tuned)



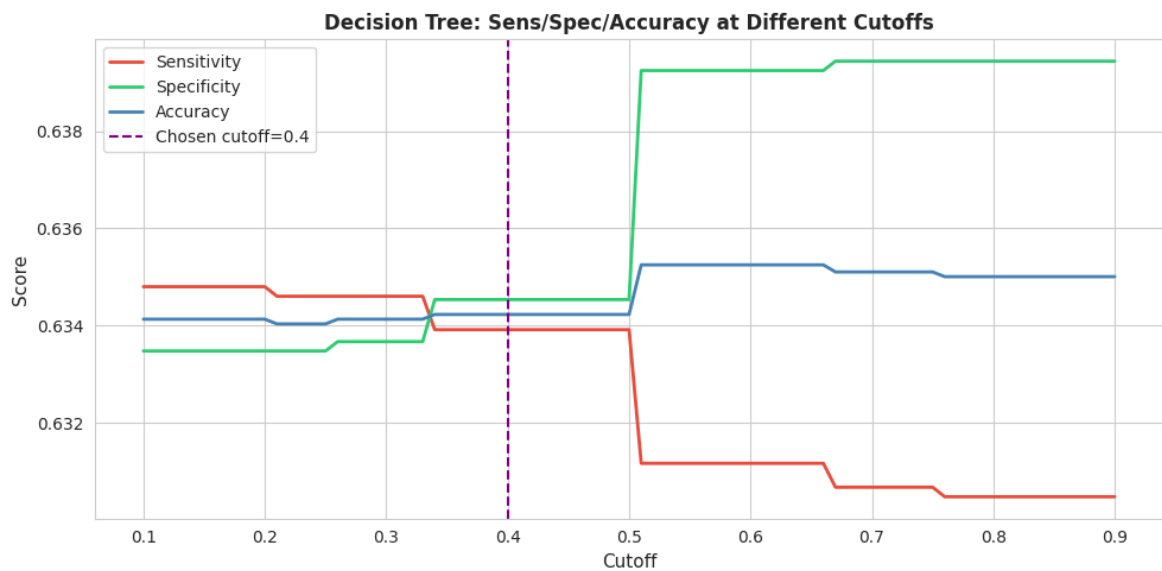
Baseline Decision Tree feature importances

Baseline DT identifies BMI, age, ap_hi as top predictors. However, overfitting makes these importances unreliable — the tree has memorised the training data.



ROC Curve — Baseline DT (AUC = 0.635)

Baseline DT: Train acc = 0.996, Test acc = 0.634. Severe overfitting confirmed. Hyperparameter tuning essential.



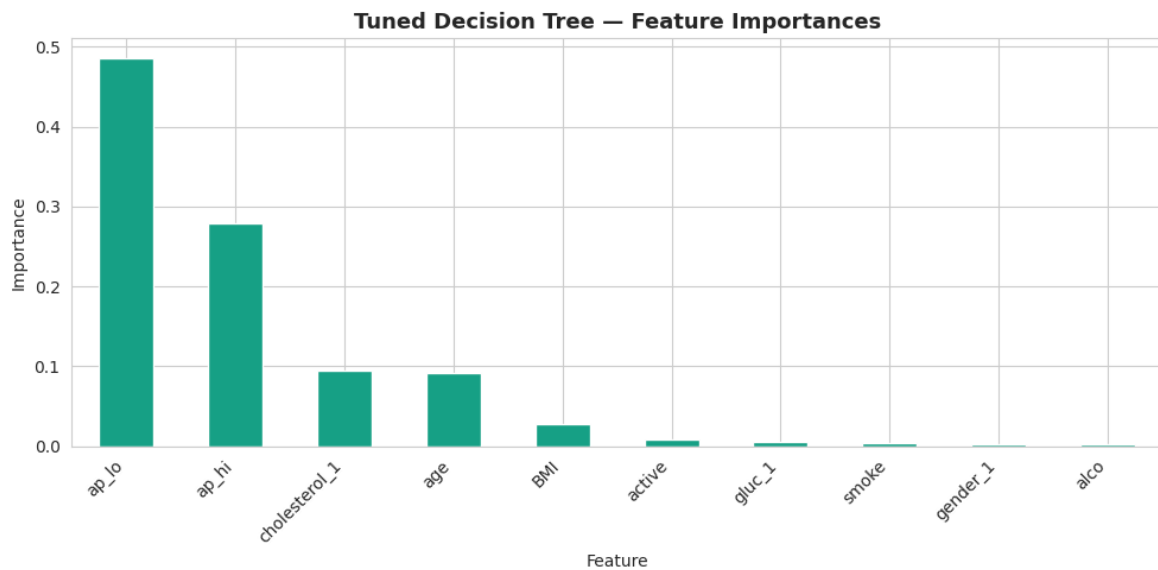
Baseline DT: Sensitivity, Specificity, Accuracy across thresholds

5.3 Hyperparameter Tuning (GridSearchCV)

5-fold GridSearchCV optimising recall. Parameters tuned: max_depth, min_samples_split, min_samples_leaf, max_features, criterion.

Parameter	Search Values	Best Value
max_depth	[3, 5, 7, 10]	10
min_samples_split	[20, 50, 100]	100
min_samples_leaf	[10, 20, 50]	20
max_features	[sqrt, log2]	sqrt
criterion	[gini, entropy]	gini

Best CV recall: **0.6884**. Constrained tree (depth=10) significantly reduces overfitting.



Tuned Decision Tree feature importances

Tuned DT shifts importance to **blood pressure** (ap_lo=0.486, ap_hi=0.279), then cholesterol_1 (0.095) and age (0.091) — clinically meaningful and consistent with EDA.

6. Model Cards

Model Card 1: Linear SVM (Production Candidate)

Attribute	Detail
Algorithm	SVC (kernel=linear, C=1.0, probability=True)
Training data	15,000-sample stratified subsample
Threshold	0.40 (optimised for recall)
Top features	ap_hi, ap_lo, age, BMI, cholesterol_1
Preprocessing	MinMaxScaler on continuous; one-hot encoding on categoricals

Split	Accuracy	Recall	Precision	AUC-ROC
Train	0.719	0.786	—	0.787
Test	0.717	0.770	0.692	0.787

- Minimal overfitting: train-test gap = 0.003
- Catches ~76% of CVD-positive patients at chosen threshold
- Limitation: subsample training; computational constraint for full SVM

Model Card 2: Decision Tree Baseline (Comparison Only)

Attribute	Detail
Algorithm	DecisionTreeClassifier (default, unconstrained)
Status	NOT recommended — severe overfitting
Threshold	0.40

Split	Accuracy	Recall	AUC-ROC
Train	0.996	~1.000	~1.000
Test	0.634	0.634	0.635

- Severe overfitting confirmed. Included only as baseline comparison.

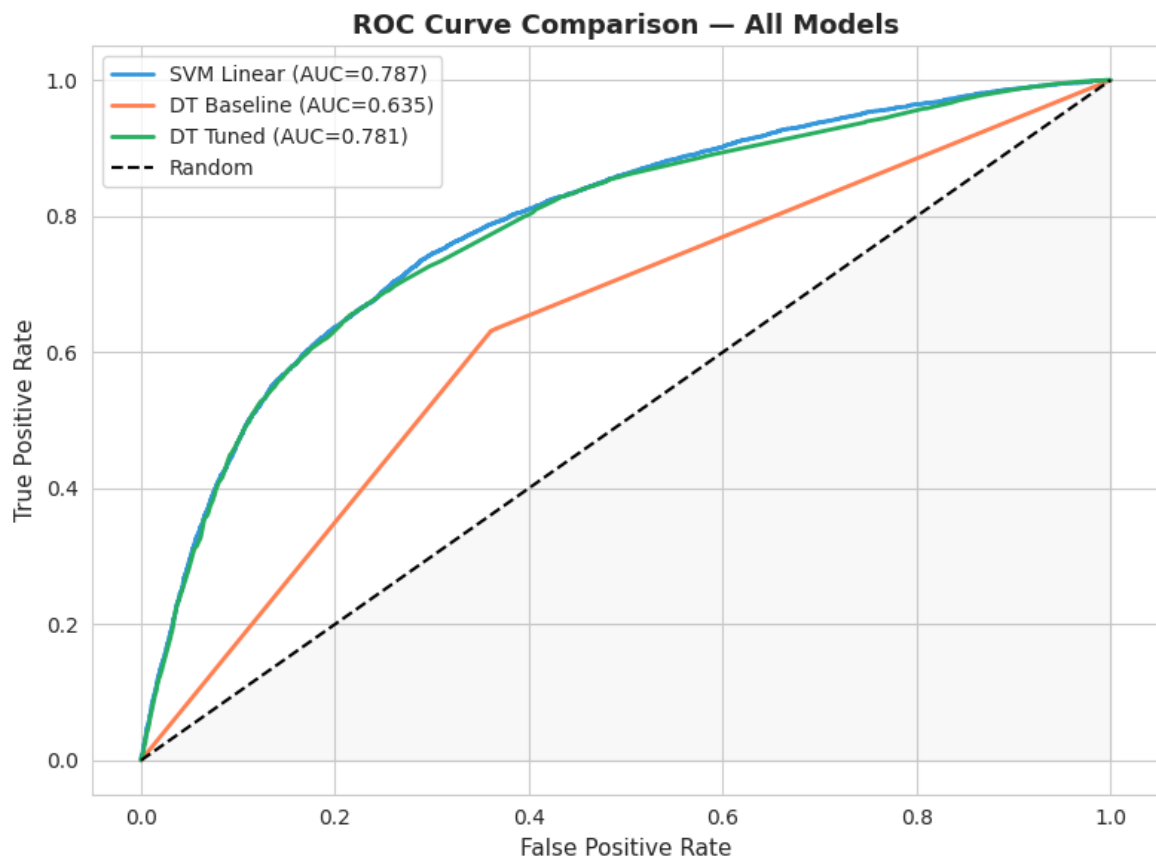
Model Card 3: Tuned Decision Tree (RECOMMENDED)

Attribute	Detail
Algorithm	DecisionTreeClassifier (tuned via GridSearchCV)
Key params	max_depth=10, min_samples_leaf=20, min_samples_split=100, criterion=gini
Training data	Full 48,052-sample training set
Threshold	0.40
Top features	ap_lo (0.486), ap_hi (0.279), cholesterol_1 (0.095), age (0.091)
Tuning	5-fold CV GridSearchCV, scoring=recall

Split	Accuracy	Recall	Precision	AUC-ROC
Train	0.711	0.816	—	0.781
Test	0.700	0.802	0.663	0.781

- Highest test recall (0.80) — catches 80% of CVD-positive patients
- Overfitting resolved: train-test gap = 0.011
- Fully interpretable tree structure; blood pressure dominates — clinically meaningful

7. Final Model Comparison



ROC curve comparison — all three models

Tuned DT and SVM achieve near-identical AUC (0.781 vs 0.787), both substantially above the baseline DT (0.635). Hyperparameter tuning is essential.

Model	Train Acc	Test Acc	Train Recall	Test Recall	Test AUC	Overfit?
SVM Linear	0.719	0.717	0.786	0.770	0.787	Minimal
DT Baseline	0.996	0.634	~1.000	0.634	0.635	Severe
DT Tuned	0.711	0.700	0.816	0.802	0.781	Minimal

8. Final Conclusion and Recommendations

8.1 Key EDA Findings

- Blood pressure (systolic and diastolic) is the strongest physiological CVD predictor
- Elevated cholesterol (76% CVD prevalence) is the most discriminating categorical feature
- Age shows a clear distribution shift in CVD-positive group
- Gender, smoking, alcohol: surprisingly weak associations — near 50% CVD rate for both categories
- Perfectly balanced target (50.5/49.5%) — no resampling required

8.2 Data Sufficiency

With 68,647 clean balanced records, the dataset is sufficient for robust modelling. AUC ~0.78-0.79 suggests meaningful signal exists, though additional clinical features (ECG, HbA1c, family history) could push AUC above 0.85.

8.3 Is a Linear Model Sufficient?

Linear SVM (AUC=0.787) and Tuned DT (AUC=0.781) achieve near-identical performance. The data has moderate non-linearity; a well-constrained DT at depth-10 captures it adequately, matching the linear SVM.

8.4 Final Model: Tuned Decision Tree (Recommended)

- **Highest recall (0.80)** — captures 80% of CVD-positive patients
- Good accuracy (0.70) and AUC (0.781)
- Interpretable tree structure — clinicians can validate decisions
- Trained on full dataset (vs SVM subsample) — maximises information
- Overfitting resolved — train-test gap reduced to 0.011

8.5 Clinical Integration Recommendations

- Deploy Tuned DT as first-line automated screening filter for new patients
- Flag patients with CVD probability ≥ 0.40 for priority physician consultation
- ~20% of CVD patients missed at this threshold — maintain secondary clinical review
- Retrain quarterly as patient demographics evolve
- Collect additional features (ECG, family history, HbA1c) to target AUC > 0.85
- Monitor calibration drift in production; recalibrate threshold annually

Report based on [CardioRisk_Prediction_Mayuresh_Tungare.ipynb](#) — available at the [GitHub repository](#).